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Physiologically Based Pharmacokinetic Modeling of Elexacaftor/Tezacaftor/Ivacaftor in Infants With Cystic Fibrosis

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ABSTRACT

Ivacaftor is the only cystic fibrosis transmembrane conductance regulator modulator approved for infants ≥ 1 month. The elexacaftor/tezacaftor/ivacaftor combination, approved for children aged ≥ 2 years, has been shown to significantly slow CF progression. Expanding elexacaftor/tezacaftor/ivacaftor use to infants may offer additional therapeutic benefits. This study aims to predict elexacaftor/tezacaftor/ivacaftor pharmacokinetics in infants aged 1 month to < 2 years using physiologically based pharmacokinetic modeling (PBPK) and to determine whether age- and weight elexacaftor/tezacaftor/ivacaftor doses, extrapolated from ivacaftor monotherapy, produce drug exposures comparable to adult levels. Adult elexacaftor/tezacaftor/ivacaftor PBPK published models were first reproduced using Simcyp. These models were scaled to children aged 2–18 years via the Simcyp Pediatric Simulator and the Upreti–Wahlström cytochrome-P450 3A4 ontogeny method to simulate the current elexacaftor/tezacaftor/ivacaftor dosing regimen. FDA-approved ivacaftor monotherapy dosing was evaluated in children 1 month to < 2 years and subsequently used to guide the extrapolation of elexacaftor/tezacaftor/ivacaftor combination dosing regimens. The overall workflow was validated using in vivo pharmacokinetic data or literature-based sources. Elexacaftor/tezacaftor/ivacaftor PBPK models for adults and children ≥ 2 years were validated against observed clinical pharmacokinetic data. Simulations of elexacaftor/tezacaftor/ivacaftor doses based on ivacaftor monotherapy in infants < 2 years matched adult-reported levels, except in infants 1–2 months, who showed lower exposures, and in those weighing ≥ 14 kg, who may experience higher exposures. Using PBPK modeling, supported by clinical pharmacokinetic data, this study explores elexacaftor/tezacaftor/ivacaftor dosing in children under 2 years. The findings offer preliminary dosing guidance, pending confirmation in larger clinical studies, with therapeutic drug monitoring recommended to guide treatment.

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Study Highlights

- What is the current knowledge on the topic?
 - Ivacaftor is the only CFTR modulator currently approved for use in children with cystic fibrosis aged 1 month to less than 2 years. The triple combination elexacaftor/tezacaftor/ivacaftor (ETI) is approved from age 2 years and was shown to significantly improve outcomes in children. Physiologically based pharmacokinetic (PBPK) modeling offers a mechanistic framework to predict drug exposure, guiding dose selection and optimization in pediatric populations with limited clinical data.
- What question did this study address?
 - This study aimed to predict the pharmacokinetics of ETI in infants aged 1 month to less than 2 years using PBPK modeling, and to determine whether age- and weight ETI doses, extrapolated from ivacaftor monotherapy, produce drug exposures comparable to adult levels.
- What does this study add to our knowledge?
 - This study simulated ETI dosing in virtual infants using regimens extrapolated from approved ivacaftor monotherapy and reports the resulting drug exposures. Simulations suggest that most infants would achieve systemic exposures comparable to those observed in adults, although variability related to age and body weight was noted.
- How might this change drug discovery, development, and/or therapeutics?
 - These findings support the rational design of pediatric dosing strategies for ETI in infants under 2 years of age. While awaiting clinical confirmation, ETI dosing extrapolated from approved ivacaftor monotherapy may provide initial guidance for early access to triple therapy in this vulnerable population, provided that treatment is initiated cautiously and progressively, with therapeutic drug monitoring and close clinical and biological follow-up.

1 | Introduction

Cystic fibrosis (CF) is an autosomal recessive inherited disease caused by mutations in the cystic fibrosis transmembrane conductance regulator (CFTR) gene [1]. These mutations result in defective CFTR protein function, disrupting ion transport across epithelial cells and thereby contributing to the characteristic pulmonary manifestations and multisystem involvement observed in CF [2]. By the age of 1 year, approximately 85% of infants with CF exhibit pancreatic insufficiency [3]. Additionally, although lung function is typically normal at the time of diagnosis through newborn screening, it declines in older infants, with many of these infants developing small airway dysfunction within the first six months of life [3–5]. Linnane et al. suggest that this period represents an optimal window for therapeutic interventions aimed at preserving lung function in CF [4].

CFTR modulators are small-molecule drugs that partially restore the function of defective CFTR proteins. The two primary classes of CFTR modulators are potentiators (e.g., ivacaftor), which enhance CFTR channel activity, and correctors (e.g., el-exacaftor, tezacaftor), which improve CFTR protein folding and trafficking. Ivacaftor is the only modulator approved for treating people with CF age 1 month to less than 2 years who have at least one CFTR mutation shown to respond to ivacaftor, based on clinical and/or in vitro assay data. The triple-combination therapy of elexacaftor-tezacaftor-ivacaftor (ETI) has received regulatory approval for children with CF (chCF) aged 2 years and older. Clinical trials have demonstrated that ETI is safe and effective, significantly improving lung function, body weight, and sweat chloride levels after 6 months of treatment in chCF aged 2 years and older, who have at least one F508del mutation [5–8]. In particular, ETI combination efficacy is superior to the tezacaftor plus ivacaftor combination, while maintaining an overall favorable safety profile [9].

Early initiation of CFTR modulator therapy has the potential to substantially alter the course of pulmonary disease in CF [6]. Indeed, a recent study suggests that initiating ETI treatment at an early stage (e.g., 2–5 years old) may lead to greater improvements in CFTR function and could potentially enhance pancreatic function in children who already exhibit documented exocrine pancreatic insufficiency [5]. This finding is consistent with earlier observation [10–12] on the use of ivacaftor monotherapy. Infants below 2 years are currently being treated [13] with ETI and clinical trials are ongoing (ClinicalTrials.gov Identifier: NCT05882357). It is expected that these therapies may prevent damage of pulmonary and pancreatic function.

However, selecting a safe and effective dose in pediatric clinical studies is challenging due to the nonlinear nature of maturation and growth, which influence physiological and biological processes. This complexity makes the selection of a safe and effective dose for pediatric clinical studies particularly challenging. Allometric scaling has been widely used to extrapolate pediatric drug dosing from adult doses based on body weight or body surface area. Furthermore, due to ongoing organ maturation, changes in body composition, ontogeny of drug metabolism and elimination pathways, simple allometric approaches often fail to accurately predict pharmacokinetic parameters in infants and neonates [14]. In response to this concern, physiologically based pharmacokinetics (PBPK) modeling, a mechanistic approach, has been used to predict adult pharmacokinetics and to extrapolate these data to pediatric populations [15]. PBPK modeling integrates compound-specific physicochemical and pharmacokinetic parameters with age-dependent physiological and anatomical data to quantitatively predict drug disposition in pediatric populations. This approach has a wide range of applications in pediatric populations, including informing initial dose selection [14–16], predicting potential drug–drug interactions, and evaluating the impact of organ impairment on pediatric pharmacokinetics [17].

The aims of this study were to predict the PK of ETI in infants aged 1 month to less than 2 years using PBPK modeling. We evaluated if age- and weight-based dosing regimens extrapolated from ivacaftor monotherapy could achieve drug exposures comparable to those observed in adults.

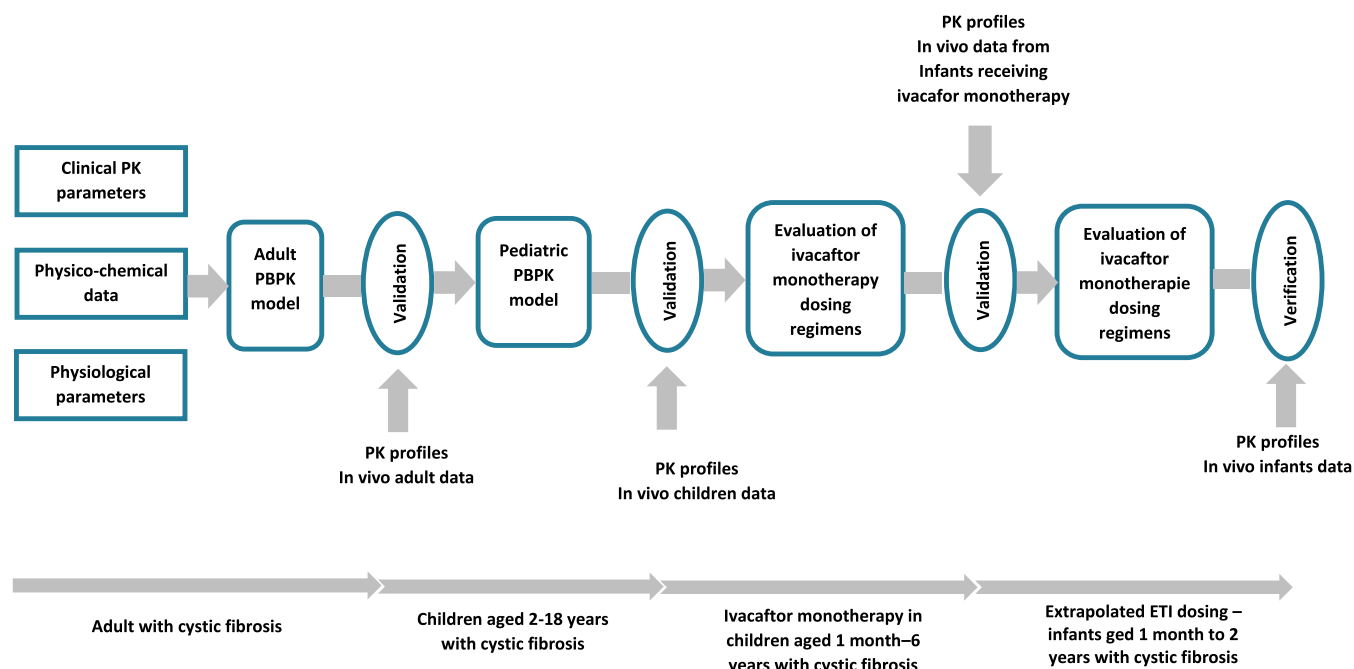


FIGURE 1 | Schematic representation of physiologically based pharmacokinetic (PBPK) model development workflow.

2 | Methods

All simulations were performed using the Simcyp PBPK Simulator version 22 (Certara, Sheffield, UK). The overall modeling workflow, covering adult model development, pediatric scaling, infant extrapolation and ETI dose evaluation, is presented in Figure 1.

2.1 | Compound Model Development

The physicochemical properties of each ETI compound, along with their absorption, distribution, metabolism, and elimination characteristics, were obtained from a previously published CF adult model [16]. The parameters used to construct the ETI compound files are detailed in File S1.

ETI combination simulations were performed by simultaneous administration of elxacaftor, tezacaftor, and ivacaftor using their respective compound files. This allowed the ETI models to capture the CYP3A-mediated interactions within the regimen, including competitive inhibition by tezacaftor and shared CYP3A metabolism of elxacaftor and ivacaftor.

Ivacaftor monotherapy simulations were performed using the ivacaftor compound file alone, without inclusion of tezacaftor and elxacaftor, to allow direct comparison with published ivacaftor monotherapy PK data.

2.2 | Virtual Populations and System Parameters

For the adult CF population, the healthy volunteer population available in the Simcyp Simulator was initially selected as the system framework, prior to the implementation of demographic

adjustments described in the Modeling workflow section. For pediatric CF patients aged 1 month to 18 years, the Simcyp pediatric population was used. Age-dependent CYP3A ontogeny was incorporated, and different maturation approaches were evaluated, as detailed in the Modeling workflow section. Virtual cohorts were defined using age and weight categories aligned with current ETI or ivacaftor dosing recommendations (Tables S1 and S2). The same age- and weight-based definitions used for ivacaftor monotherapy were applied to extrapolate ETI dosing in patients aged under 2 years (Table S3).

2.3 | Modeling Workflow

2.3.1 | Adult ETI PBPK Models

For adult PBPK modeling, the virtual healthy volunteer population in the Simcyp Simulator was adjusted by increasing the proportion of females from 32% to 50% to better reflect the adult cystic fibrosis population [16–18]. Dose simulations were conducted using current adult ETI dosing recommendations in 35 trials of 10 subjects each (350 virtual subjects). The simulated age and weight demographic characteristics were consistent with those reported in the adult clinical datasets used for model development and validation [18, 19]. Adult ETI pharmacokinetics were simulated under fed conditions to reflect clinical administration requirements. Steady-state conditions were evaluated following 14 days of repeated dosing. Model performance was assessed by visual inspection of predicted versus observed plasma concentrations, using therapeutic drug monitoring data collected from adults with cystic fibrosis in two independent clinical studies [18, 19] (Table S4). Additionally, the proportion of observed concentrations falling within the 90% prediction interval of the simulated data across adult populations was evaluated [15].

2.3.2 | Pediatric ETI PBPK Models (2 to 18 Years)

The validated adult ETI PBPK models were then scaled to children aged 2 to <18 years. The Simcyp pediatric virtual population was used without further adjustment to the age–weight distribution, following comparison with clinical data from toddlers and preschool-aged children with CF enrolled in the MODUL-CF study (ClinicalTrials.gov Identifier: NCT04301856) (see [Supporting Methods](#)), which revealed no meaningful differences (Table S5). The sex ratio was adjusted to 1:1. Two different CYP3A ontogeny maturation approaches were evaluated as described in the [Supporting Methods](#), and the Upreti and Wahlstrom approach was finally selected. Current pediatric ETI recommended doses (Table S1) were simulated over 14 days to reach steady state exposure levels. Approximately 200 virtual subjects per age and weight ETI drug recommendation were simulated. Model performance and accuracy were evaluated using data from the MODUL-CF study [20] (Table S4) by: (1) visual inspection of predicted versus observed ETI plasma concentration–time profiles for each age- and weight-based dosing recommendation, including assessment of the proportion of observed concentrations falling within the 90% prediction interval; and (2) comparison of steady-state PK parameters, including area under the curve (AUC), maximal concentration (C_{max}), and minimal concentration (C_{min}), with reported observed values; and calculation of quantitative metrics, including mean fold error (MFE), average fold error (AFE), and absolute average fold error (AAFE). A full description of the evaluation metrics is provided in the model evaluation criteria section.

2.3.3 | Infant Ivacaftor PBPK Model (1 Month–<2 Years)

The ivacaftor compound adult PBPK model was scaled to infants aged 1 month to 2 years. The sex ratio of the Simcyp pediatric virtual population was adjusted to 1:1, while the age–weight distribution remained unchanged. Ivacaftor monotherapy was simulated using the current recommended dosing regimens (Table S2), with approximately 200 virtual subjects simulated per regimen over a 14-day dosing period. For evaluation of infant ivacaftor PBPK model performance and accuracy, only literature-based data were available (Table S4), reporting AUC values. Accordingly, simulated steady-state AUC_{0-12h} values for each dosing regimen were compared with label-reported infant ivacaftor exposure data [21]. Predictions for infants aged 1 to <4 months were additionally compared with data reported by McNally et al. focusing on the youngest age group among the seven infants with cystic fibrosis included in that study (File S2). Model performance was assessed using MFE, AFE, and AAFE metrics.

For interpretative purposes, the predicted AUC_{0-12h} values were additionally compared with adult reference exposures from monotherapy studies [21].

2.3.4 | Infant ETI PBPK Models and Dose Extrapolation (1 Month–<2 Years)

Pediatric ETI PBPK models were subsequently scaled to children under 2 years of age. Model performance was initially

assessed by visual inspection using real-world data from three infants (File S3), with approximately 100 simulations performed to evaluate predictive performance based on the dosing received by these children and matched age and weight ranges.

We then derived ETI predictions by extrapolating elexacaftor and tezacaftor dosing from ivacaftor monotherapy. The evening ivacaftor monotherapy dose was preserved, while the morning ivacaftor dose was used to calculate the corresponding elexacaftor and tezacaftor doses, maintaining the standard component ratio. Fixed drug content per granule (72 granules per ETI 80/40/60 mg sachet; 31 granules per ivacaftor 59.5 mg sachet) enabled accurate individualization of doses through granule counting. Simulations of approximately 200 virtual subjects per regimen were performed using the Simcyp pediatric virtual population over 14 days to estimate steady-state exposure. For interpretative purposes, predicted $AUC_{0-\tau}$ values were compared with adult reference exposures [6] while predicted ETI C_{max} values were evaluated against the 95% confidence intervals reported in the literature [22, 23].

Additionally, the predicted average plasma concentrations of tezacaftor and elexacaftor, along with the minimum plasma concentrations of ivacaftor, were plotted against literature-derived total EC_{50} and EC_{90} thresholds [16]. The total EC_{50} values for tezacaftor and ivacaftor were originally derived by previous teams [16, 24] from clinical exposure–response analyses based on sweat chloride reduction and were not determined in combination with elexacaftor [22, 23, 25]. In contrast, the total EC_{50} value for elexacaftor was derived from clinical studies conducted in combination with tezacaftor/ivacaftor, using efficacy endpoints including sweat chloride and percent predicted forced expiratory volume in 1 s [23].

All EC_{50} and EC_{90} values applied in this analysis reflect total plasma concentrations, as no data on efficacy relationships with unbound concentrations are currently available. The values used in the present study were extracted from Tsai et al. using PlotDigitizer (<https://plotdigitizer.com/app>) and were applied as contextual reference ranges rather than definitive pharmacokinetic–pharmacodynamic targets, to support the assessment of dose adequacy in infants.

2.4 | Model Evaluation Criteria

Model accuracy in adults, children, and infants was evaluated using one or more of the following methods, depending on the availability of validation data: [1] visual inspection of observed versus simulated concentrations; [2] comparison of steady-state pharmacokinetic parameters; and [3] quantitative metrics, including MFE, AFE, and AAFE. These quantitative metrics were used to compare predicted and observed pharmacokinetic parameters using predefined calculation formulas.

The MFE was calculated for C_{max} , C_{min} , and $AUC_{0-\tau}$ PK parameters according to the following formula:

$$MFE = \frac{\text{Predicted mean parameter}}{\text{Observed mean parameter}}$$

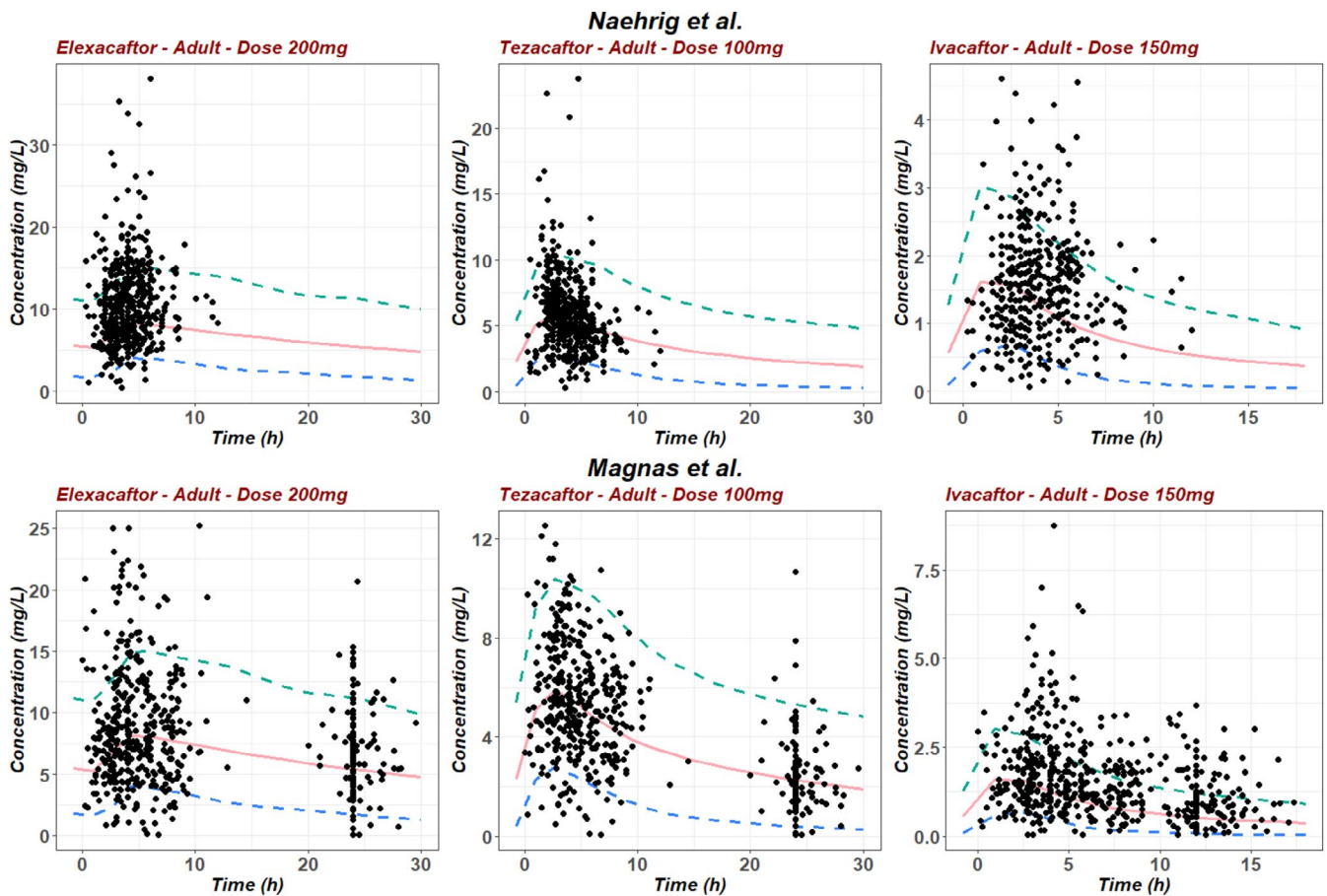


FIGURE 2 | Validation graphs of elexacaftor, tezacaftor, and ivacaftor according to adult dosing recommendations. Solid lines indicate mean simulated concentrations; blue and green dotted lines represent the 5th and 95th percentiles, respectively. Dots represent observed ETI data. Upper panels show data from Naehrig et al., and lower panels show data from Magnas et al. [18, 19].

The MFE was classified within the predefined error range of 0.5–2, representing an acceptable two-fold error margin for model validation [26, 27].

Overall model bias and precision were further assessed using the AFE and AAFE, with particular emphasis on the MFE for drug exposure quantified by $AUC_{0-\tau}$, as follows:

$$AFE = 10^{\frac{1}{n} \sum \log_{10} \left(\frac{\text{Predicted mean parameter}}{\text{Observed mean parameter}} \right)}$$

$$AAFE = 10^{\frac{1}{n} \sum |\log_{10} \left(\frac{\text{Predicted mean parameter}}{\text{Observed mean parameter}} \right)|}$$

The AFE reflects overall model bias, indicating systematic over- or under-prediction across simulations, whereas the AAFE describes overall predictive precision irrespective of direction. An AAFE within two-fold was considered acceptable, while an AAFE within 1.25-fold was interpreted as good model performance [28].

3 | Results

3.1 | Adult ETI PBPK Models

The adult PBPK models for elexacaftor/tezacaftor/ivacaftor were qualified by comparing the median, 5th, and 95th percentile predicted concentration–time profiles with observed

pharmacokinetic data from the literature [18, 19], ensuring their suitability for extrapolation to pediatric populations. The validation graphics are presented in Figure 2, and Table S6A illustrating the agreement between predicted and observed adult ETI plasma concentration–time profiles. Across adult datasets, 72%–92% of observations fell within the 5th–95th prediction interval depending on the compound, with the best performance observed for tezacaftor.

3.2 | Pediatric ETI PBPK Models (2 to 18 Years)

The adult PBPK models were scaled to children aged 2 to 18 years, covering a broad range of body weights. Predicted ETI plasma concentration–time profiles for each recommended dosing regimen (Table S1) showed good agreement with observed data from the MODUL-CF study, as illustrated in Figure 3 and Table S6B. Across age and weight groups, 70%–100% of observed concentrations fell within the 90% prediction interval, with a greater dispersion observed in younger children and for ivacaftor.

Model performance was further assessed using the MFE for $AUC_{0-\tau}$, $C_{\max}$, and $C_{\min}$ across all compounds and dosing regimens (Table 1, Figure S1). For all age and weight groups, MFE values for ETI were within the predefined acceptance range of 0.5–2. Prediction accuracy was generally highest for $AUC_{0-\tau}$,

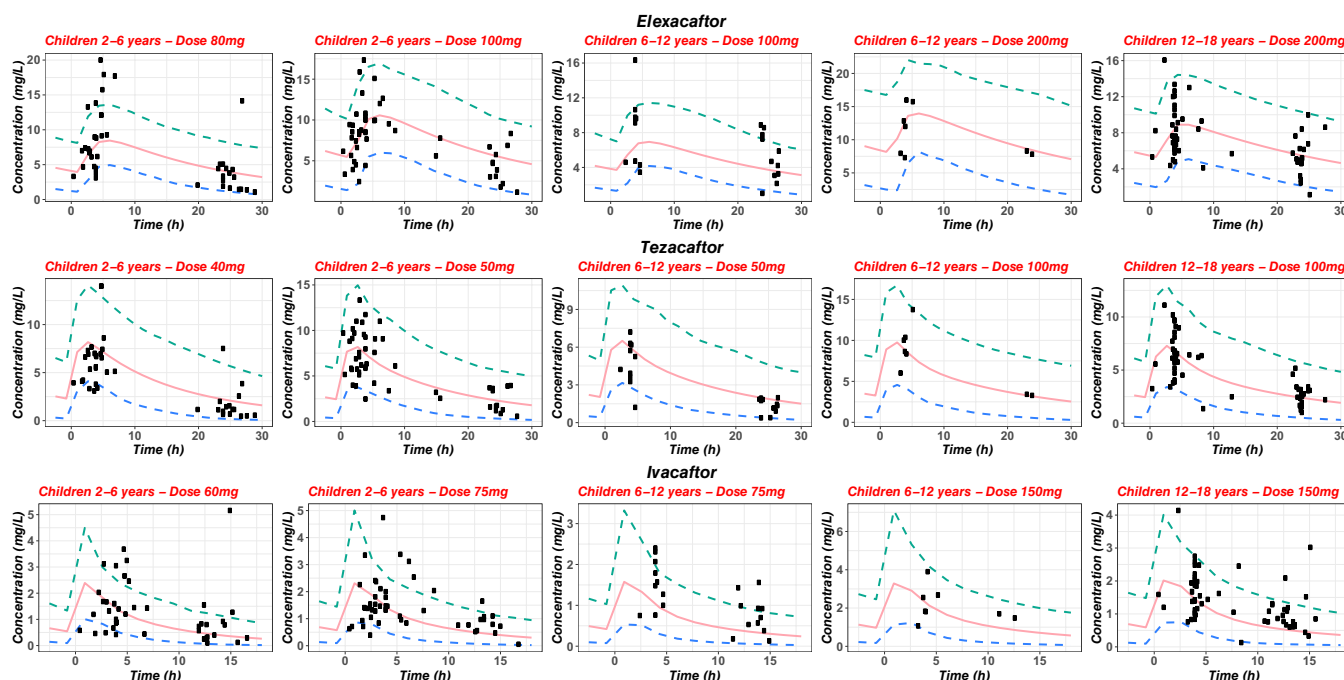


FIGURE 3 | Validation graphs of elexacaftor, tezacaftor, and ivacaftor as a function of dosing recommendations in children aged 2 to 18 years, stratified by weight and age groups. Solid lines indicate mean simulated concentrations; green dotted lines represent the 5th and 95th percentiles. The dots stand for ETI observed data [20].

whereas $C_{\max}$ and $C_{\min}$ exhibited greater deviations, particularly for ivacaftor.

Overall model bias and precision for $AUC_{0-\tau}$ were satisfactory for all ETI components, with a range of AFE/AAFE values of 0.98/1.09 for elexacaftor, 1.00/1.07 for tezacaftor, and 0.89/1.13 for ivacaftor. Taken together, both graphical and numerical evaluations support good predictive performance of the ETI PBPK models in children aged 2 years and older.

3.3 | Infant Ivacaftor PBPK Model Verification

The predictive performance of the ivacaftor PBPK model was evaluated in infants younger than 2 years of age. Simulations were conducted using FDA-recommended monotherapy dosing regimens (Table S2), and simulated AUC_{0-12h} values are shown as a function of age and body weight in Figure 4. Ivacaftor doses administered in the study by McNally et al. in children with CF aged <4 months were also simulated (File S2). Across all age and weight groups and dosing regimens, MFE values for ivacaftor $AUC_{0-\tau}$ remained within the predefined 0.5–2-fold range (Figure 5), with overall AFE and AAFE values of 1.30 and 1.40, respectively, indicating satisfactory predictive performance. As shown in Figure 4, simulated exposures in infants aged 1 to <2 months were marginally lower than those observed in older infants, indicating lower ivacaftor exposure in the youngest age group. In contrast, a tendency toward overprediction was noted in children older than 4 months, especially in infants aged 4 to 12 months, although most simulated values remained within the adult 5th–95th percentile range.

3.4 | Infant ETI PBPK Modeling and Extrapolation of ETI Dosing in Infants Aged 1 Month to 2 Years

The performance of the ETI PBPK models in children under 2 years of age was evaluated using data from three infants (aged 7–16 months) enrolled in the MODUL-CF study, showing good agreement between simulations and observations (File S3). Based on this initial validation and the satisfactory predictive performance of the ivacaftor PBPK model in infants, PBPK-informed dosing regimens for elexacaftor and tezacaftor were simulated in infants aged 1 month to <2 years, using FDA-recommended ivacaftor monotherapy as the reference. The resulting steady-state ETI exposures are presented in Figure 6 and Table S7.

Simulated $AUC_{0-\tau}$ values were lower than adult reference exposures in the youngest infants (Figure 6). In contrast, infants aged 6 months to <2 years and weighing ≥ 14 kg exhibited higher $AUC_{0-\tau}$ values for elexacaftor and tezacaftor than adults, whereas ivacaftor exposures remained within the adult range. Elxacaftor exhibited the greatest inter-regimen variability in AUC_{0-t} , while tezacaftor and ivacaftor exposures largely matched adult targets.

Simulated ETI $C_{\max}$ values are shown in Figure S2. In infants weighing >14 kg, $C_{\max}$ tended to exceed adult reference ranges, particularly for ivacaftor, a trend already observed in the 7–14 kg group. In the youngest infants, elxacaftor and tezacaftor $C_{\max}$ values were generally lower than adult references, while ivacaftor values remained closer to adult levels.

Figure S3 shows the simulated C_{avg} of elexacaftor and tezacaftor, and the $C_{\min}$ of ivacaftor, compared to previously reported EC_{50}

TABLE 1 | Comparison of predicted PK parameters (PBPK Models) and reference PK parameters at steady state of children aged 2 years and older [21].

Group	AUC0_tho		AUC0_tho reference	MFE	C _{max} prediction	C _{max} reference	MFE	C _{min} prediction	C _{min} reference	MFE
Mean (sd) PK parameters of elexacaftor										
12 to <18 years		171.7 (65.8)	168 (51)	1.02	9.1 (3.0)	8.7 (2.3)	1.04	5.1 (2.5)	5.1 (1.9)	1
6 to <12years., WT ≥ 30kg		266.6 (108.5)	233 (33)	1.14	14.2 (4.7)	12.2 (1.5)	1.16	7.8 (4.3)	6.9 (1.2)	1.13
6 to <12years., WT <30kg		129.1 (48.8)	154 (53)	0.84	7.0 (2.24)	8.3 (2.2)	0.84	3.6 (1.8)	4.4 (2.0)	0.82
2 to <6years., WT ≥ 14kg		193.1 (85.1)	187 (53)	1.03	10.7 (3.8)	10.3 (2.2)	1.04	5.2 (3.3)	5.1 (2.0)	1.02
2 to <6years., WT <14kg		149.6 (59.9)	162.0 (32)	0.92	8.6 (2.8)	9.6 (1.3)	0.90	3.8 (2.3)	3.8 (1.2)	1
Mean (sd) PK parameters of tezacaftor										
12 to <18 years		103.1 (47.1)	102 (21)	1.01	7.4 (3.0)	6.2 (1.2)	1.19	2.4 (1.5)	2.4 (0.7)	1.0
6 to <12years., WT ≥ 30kg		137.2 (74.8)	139 (23)	0.99	10.1 (4.1)	8.61 (1.1)	1.17	3.2 (2.6)	3.2 (0.7)	1.0
6 to <12years., WT <30kg		91.4 (42.1)	78 (16)	1.17	6.7 (2.5)	5.4 (0.8)	1.24	1.96 (1.4)	1.43 (0.5)	1.37
2 to <6years., WT ≥ 14kg		113.6 (59.6)	111 (23)	1.02	8.2 (3.4)	7.5 (1.1)	1.09	2.3 (2.0)	2.1 (0.8)	1.1
2 to <6years., WT <14kg		114.8 (53.5)	101 (24)	1.14	8.4 (3.0)	7.5 (1.01)	1.12	2.2 (1.8)	1.6 (0.8)	1.37
Mean (sd) PK parameters of ivacaftor										
12 to <18 years		13.8 (6.58)	15.9 (6.2)	0.87	2.2 (1.17)	1.6 (0.5)	1.37	0.55 (0.4)	1.00 (0.5)	0.55
6 to <12years., WT ≥ 30kg		18.8 (10.7)	21.0 (5.3)	0.90	2.9 (1.6)	2.1 (0.5)	1.38	0.8 (0.7)	1.3 (0.4)	0.62
6 to <12years., WT <30kg		10.8 (5.32)	13.5 (4.3)	0.80	1.9 (0.9)	1.4 (0.4)	1.36	0.40 (0.35)	0.78 (0.33)	0.51
2 to <6years., WT ≥ 14kg		14.2 (8.2)	16.2 (5.7)	0.88	2.5 (1.3)	1.7 (0.5)	1.47	0.52 (0.5)	0.89 (0.45)	0.58
2 to <6years., WT <14kg		14.07 (7.5)	13.6 (4.0)	1.03	2.6 (1.2)	1.5 (0.3)	1.70	0.5 (0.5)	0.65 (0.3)	0.77

Note: The reference PK parameters were derived from internal data, partially published in a population pharmacokinetics study of ETI involving a pediatric population aged 2 to 18 years. Abbreviations: AUC0_tho, Area under the curve (AUC_{0-12h} for ivacaftor; AUC_{0-24h} for elexacaftor and tezacaftor); C_{max}, peak concentration; C_{min}, minimal concentration; MFE, mean fold error; PK, pharmacokinetic; sd, standard deviation; WT, weight.

Simulated ivacaftor exposure based on the FDA monotherapy dosing recommendations

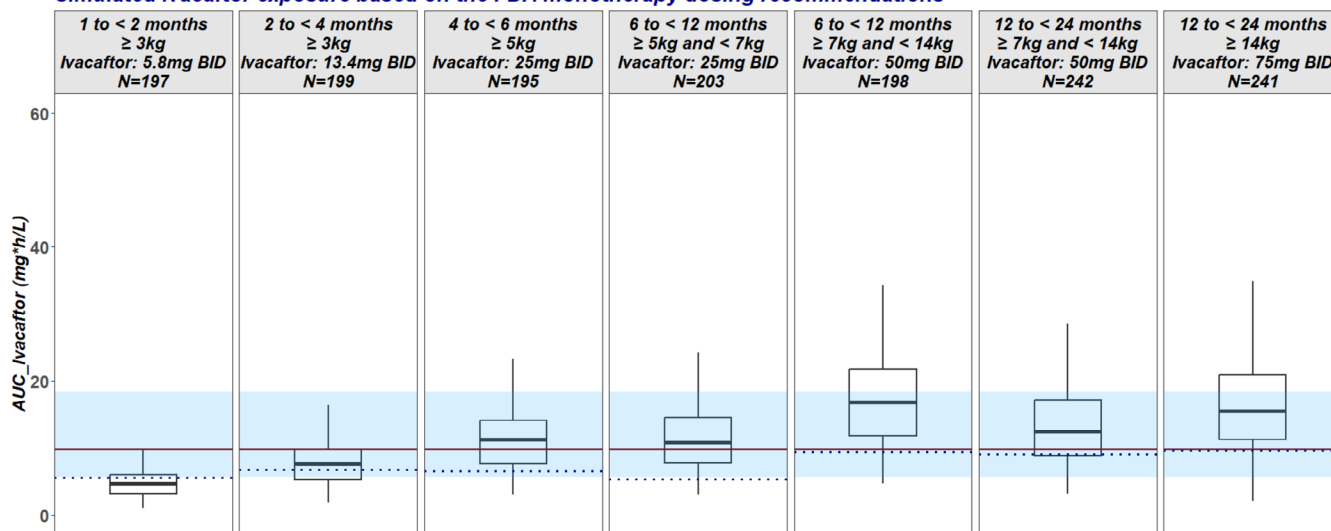


FIGURE 4 | Boxplots of steady-state ivacaftor exposure (AUC_{0-12h}) across age-weight bands simulated using the final PBPK model under current FDA monotherapy dosing recommendations. The solid line represents the reported adult median exposure, and the blue area indicates the corresponding adult 90% confidence interval [21]. The dashed line represents the observed median exposure in infants for each dosing recommendation [10, 11]. N denotes the number of simulated individuals per group.

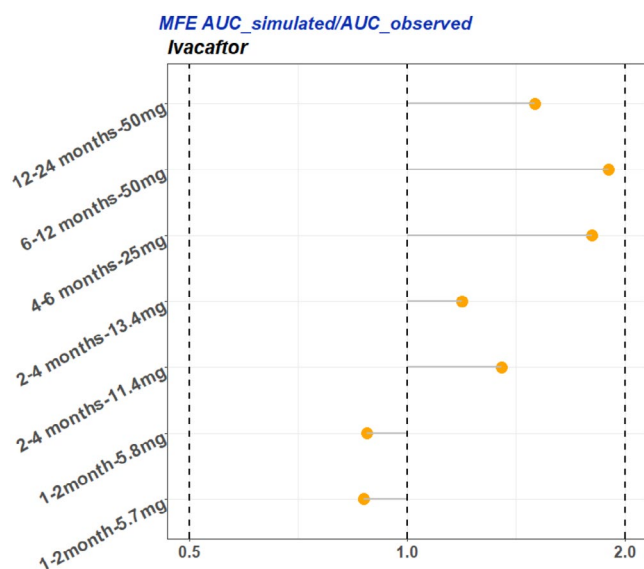


FIGURE 5 | Predicted-to-observed steady state $AUC_{0-\tau}$ ratios (MFE) for ivacaftor across dosing regimens and clinical datasets [10–12] in infants aged 1 to 24 months.

and EC_{90} values [16]. Elexacaftor C_{avg} values remained above EC_{50} in all groups, with a high proportion reaching EC_{90} at older ages and higher doses. Tezacaftor C_{avg} exceeded EC_{50} in most groups, though several infants in the youngest category were below this threshold; the proportion reaching EC_{90} increased steadily with age. Regarding ivacaftor, the percentage of C_{min} values above EC_{90} increased consistently across age groups, reflecting rising exposure from early infancy to 2 years of age.

4 | Discussion

This study aimed to extrapolate CF adult PBPK models in infants, particularly those aged 1 month to 2 years, for whom no

ETI dosing recommendations currently exist, providing the first pharmacokinetic predictions for this combination therapy in this population. ETI is predominantly metabolized by CYP3A4 with limited renal excretion, making the selection of an appropriate CYP3A4 ontogeny model for accurate pharmacokinetic predictions. After evaluating the CYP3A4 ontogeny models of Salem and Uprety-Wahlström in chCF aged 2 to 18 years, the latter demonstrated superior predictive accuracy for all three compounds. This model produced pharmacokinetic profiles that better aligned with observed data in young children and were consistent with findings from other pediatric studies involving drugs primarily metabolized by CYP3A4 [29–31]. These results emphasize the need to consider CYP3A ontogeny and maturation-related physiological changes for accurate PK predictions in infants. All quantitative performance metrics fell within the acceptable 0.5–2.0-fold error range for PBPK modeling, with $AUC_{0-\tau}$ showing particularly high accuracy across all molecules and age groups. The ivacaftor PBPK model showed a tendency to underpredict the highest observed concentrations (with over 5% of observations above the 95th percentile), suggesting that interindividual pharmacokinetic variability may not be fully captured, while MFE values were generally > 1 for C_{max} and < 1 for C_{min} .

The predictive performance of the ivacaftor PBPK model was then evaluated in children aged 1 month to 2 years by simulating ivacaftor monotherapy according to FDA-recommended age- and weight-based dosing regimens and by comparing the predictions with clinical studies data [10–13]. A trend toward overprediction of ivacaftor exposure was observed in children aged ≥ 4 months (i.e., MFE values > 1). Accurately capturing the full kinetic profile in infants, particularly the absorption phase and oral bioavailability, remains challenging in the absence of detailed in vivo data. In addition, the limited sample sizes of infant pharmacokinetic studies may further amplify apparent mispredictions due to high interindividual variability and the sensitivity of median exposure estimates. Nevertheless, model

Simulation of ETI exposure in infants aged 1 month to less than 2 years

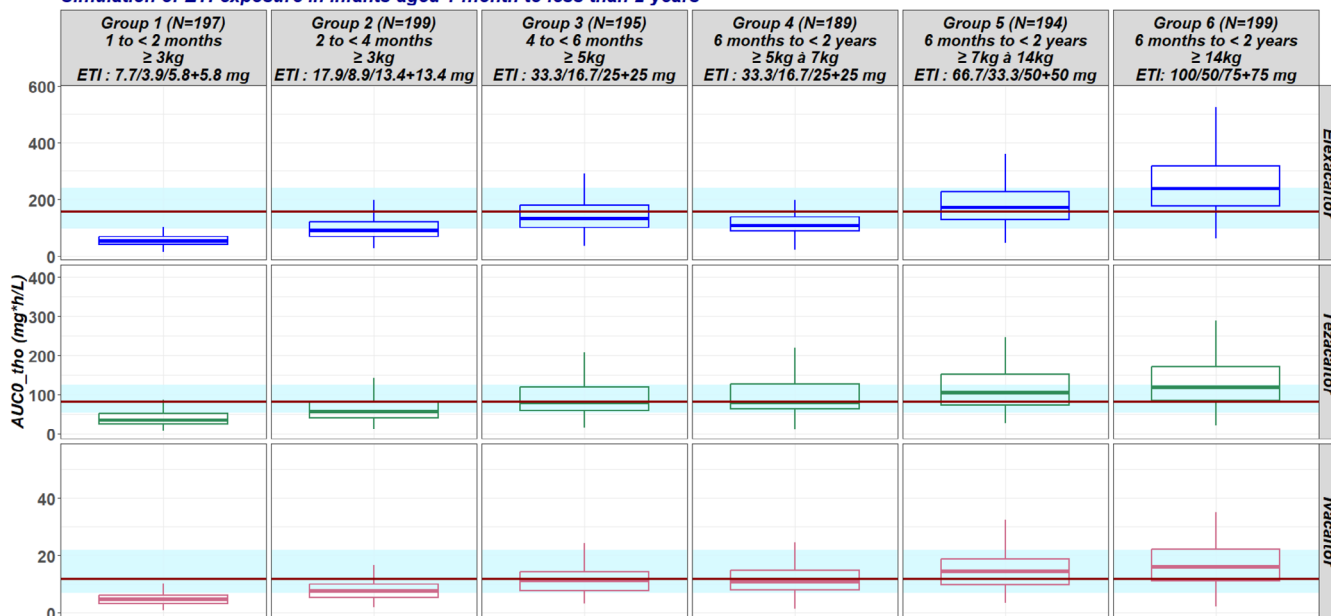


FIGURE 6 | Boxplots of the simulated steady-state ETI drug exposure (AUC_{0-tho}) across age-weight bands. The ETI drug dosing are extrapolated from the ivacaftor monotherapy FDA recommendations. The solid line represents the reported adult median exposure, and the blue area indicates the 5th and 95th percentiles of the adult AUC expected values [6]. N denotes the number of simulated individuals per group.

predictions remained within the predefined acceptable MFE range, and overall predicted AUC_{0-12h} values were consistent with those reported in the prescribing information [21].

For infants aged 1 to 4 months, an additional source of data was available from the study by McNally et al. [12], in which infants received lower doses than those recommended by the FDA for comparable age and weight groups. McNally et al. reported that AUC_{0-12h} values in these infants were lower than those observed in adults and older infants [10, 11, 21], while the predicted C_{min} exceeded the reported EC_{50} of ivacaftor [16] for all infants. Interestingly, after 2 weeks of treatment, the mean sweat chloride concentration dropped to approximately 30 mmol/L, with all valid results below 60 mmol/L. They suggest that ivacaftor treatment in infants leads to rapid and clinically meaningful improvement in CFTR function, comparable to effects seen in older children, adolescents, and adults. Thus, despite lower AUC_{0-12h} values observed in the youngest infants compared to expected adult values with CF, maintaining C_{min} above the EC_{50} may be sufficient to achieve therapeutic efficacy.

Given the potential benefits of early ETI initiation to enhance CFTR function, improve lung outcomes, prevent early lung damage, and support pancreatic function [5, 10–12], we conducted pharmacokinetic simulations to extrapolate appropriate ETI dosing in infants aged 1 month to < 2 years. Ivacaftor monotherapy was used as the starting point for defining ETI dosing in infants that could be implemented in clinical practice using ETI granule formulations, with dose individualization achieved by adjusting the number of granules administered to enable practical and accurate dosing.

Based on ETI dosing simulations, infants older than 2 months had $AUC_{0-\tau}$ values within the 5th–95th percentile range observed in adults. In contrast, in the youngest group, the $AUC_{0-\tau}$

of ETI is lower than the adult reference values. Nevertheless, the predicted C_{avg} of elexacaftor and tezacaftor, as well as the C_{min} of ivacaftor, remained above their respective EC_{50} thresholds previously reported [16]. These results, in conjunction with findings from ivacaftor monotherapy in this age group, suggest that the extrapolated ETI doses may provide adequate therapeutic exposure in infants while minimizing overall exposures.

Conversely, the simulated $AUC_{0-\tau}$ and C_{max} values in the heaviest group were high, raising concerns regarding potential higher exposure in this subgroup. Given the absence of real-world safety and tolerability data particularly in infants under 2 years of age receiving ivacaftor monotherapy at 75 mg BID for extended periods [10, 11]; and the growing number of reports of behavioral and sleep disturbances following ETI initiation in preschool-age chCF [32], careful consideration of extrapolated dosing predictions remains essential. We suggest that infants weighing ≥ 14 kg may benefit from an ETI dose lower than 100/50/75 + 75 mg.

The ETI PBPK models were evaluated using in vivo data from three infants under 2 years old who received varying doses. Despite limited data, the models reliably predicted drug exposure across infants of different ages and weights, supporting their overall predictive performance in this population. The predicted concentrations aligned well with the observed concentration-time profiles of ivacaftor and tezacaftor. For elexacaftor, the predicted absorption phase was consistent with observed concentrations in two infants; however, one infant exhibited a deviation at the four-hour post-dose measurement. Despite this discrepancy, observed concentrations during the elimination phase remained within the 5th to 95th percentile of predicted values for this child. Given the PBPK model's strong predictive accuracy for elexacaftor in children aged 2 to 18 years, the inconsistencies observed in infants likely reflect additional

factors beyond physiological maturation. One such factor may be feeding, as elexacaftor's $AUC_{0-\tau}$ is known to increase 1.9- to 2.5-fold when taken with a moderate-fat meal. However, the impact of infant feeding on ETI absorption remains poorly understood due to limited data, which could affect the accuracy of this PBPK model. Moreover, the adult PBPK models for elexacaftor and ivacaftor also include biliary clearance, which is scaled in Simcyp to account for developmental increases in liver function in the corresponding pediatric PBPK models. According to Johnson et al., biliary clearance pathways and transporter activities mature rapidly during early infancy, with timing that may vary depending on the drug and transporter involved [33, 34]. However, the specific transporters mediating the biliary clearance of elexacaftor and ivacaftor have not yet been clearly identified. In the current models, biliary clearance is implemented using adult reference values without any age-specific adjustments for overall maturation or transporter ontogeny. Simulations begin after 1 month of age, when biliary clearance is generally expected to approximate adult levels. Consequently, although uncertainty regarding transporter ontogeny remains, this is unlikely to have a major impact on predicted exposures in pediatric models. Nonetheless, a more comprehensive investigation of this clearance pathway is needed to enhance the accuracy of predicting elexacaftor and ivacaftor concentrations in young infants.

This study has several important limitations that must be acknowledged. The most important limitation is that the infant ETI PBPK models were validated using observed data from only three infants aged 7 to 16 months. This limited dataset constrains full evaluation of the model's predictive performance in infants, particularly in the youngest age group, for which clinical data are essentially lacking. Another limitation is that the model did not incorporate population-specific system parameters for adults and chCF. Disease-related changes, including gastrointestinal function, plasma protein levels, and potential differences in clearance, were not incorporated into the models. Notably, the inclusion of plasma protein levels as a CF-specific parameter remains debated [35]. Nevertheless, with recent therapeutic advances, BMI in pwCF is now comparable to that of healthy volunteers [17], and CYP3A enzyme activity appears unaffected in chCF [36, 37]. Additionally, weight distributions in toddler and preschool populations also aligned with Simcyp virtual populations and adult clinical data [38, 39]. These observations generally support the use of PBPK-based dose extrapolations, while some residual uncertainty remains for infants and very young children due to limited physiological data. A third limitation is the reduced predictive accuracy of the ivacaftor PBPK model in children and infants, highlighting the need for caution when interpreting ETI simulation results. Finally, the evaluation of ETI exposure at the extrapolated doses was primarily guided by adult reference targets, despite the absence of clearly defined PK/PD relationships in both adults and children over 2 years. This limitation may affect the determination of appropriate dosing needed for infants under 2 years to achieve the same therapeutic benefit as older children and adults.

Further investigation is needed to evaluate the therapeutic efficacy of the extrapolated ETI dosing regimen in infants aged 1 month to 2 years. While these extrapolations offer a valuable starting point for informing dose selection, clinical trial data

and therapeutic drug monitoring remain essential to ultimately guide safe and effective dosing. Given that most simulations were conducted under steady-state conditions, ETI initiation in infants should proceed gradually, with careful clinical and therapeutic drug monitoring to ensure safety, guide dose adjustments, and account for variability in drug exposure within this highly heterogeneous and vulnerable population.

This is the first study to characterize the clinical pharmacokinetics of ETI in infants using a PBPK modeling approach. Various ETI dosing regimens were evaluated by extrapolating from the recommended ivacaftor monotherapy dosage, providing an initial reference point for ETI dose selection and optimization in this age group. Despite the lack of guideline-based dosing regimens for infants aged 1 month to less than 2 years, the initiation of ETI therapy—guided by therapeutic drug monitoring and close monitoring of biological parameters—may offer clinically meaningful therapeutic benefits. While these extrapolations offer a valuable framework for early dosing decisions, the lack of a well-defined PK–PD relationship for ETI in infants underscores the exploratory nature of these findings and highlights the need for further clinical investigation.

Author Contributions

N.H.T., F.F., S.B., J.-M.T., and I.S.-G. designed and performed the research; N.H.T., F.F., S.B., J.-M.T., I.S.-G., E.D., A.M., P.R., M.R., and P.-R.B. performed the research; N.H.T., F.F., S.B., N.B., L.F.-B., S.R., M.R.S., S.U., and G.L. analyzed the data; N.H.T., F.F., S.B., and N.B. wrote the manuscript.

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The authors have nothing to report.

Conflicts of Interest

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Data S1:** psp470225-sup-0001-supinfo.zip.